Genomics and Bioinformatics Unit
Department of Infectious Diseases

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MCBBi | NOVA
UNIVERSIDADE NOVA DE LISBOA

Ministério da Saúde

Instituto_Nacional de Saúde
Doutor Ricardo Jorge

Instituto Nacional de Saúde Dr. Ricardo Jorge

State Laboratory (NIH)

- Under the Ministry of Health
- Broad mission towards the improvement of public health

Divided in several departments

- Departament of Food and Nutrition
- Departament of Infectious Diseases
- Departament of Epidemiology
- Departament of Human Genetics
- Departament of Health Promotion and Prevention of Non-Transmissible Diseases
- Departament of Environmental Health

Housing the National Reference Laboratories

Laboratory surveillance of infectious diseases

Priorities

- **Detection** of infectious agents
- **Discrimination** and **classification** (typing techniques)

Benefits for public health:

- **Tracking the emergence and spread of pathogenic agents**
- **Detect and solve outbreaks**
- **Identify factors that contribute to the persistence of endemic clones**
- **Complement/Support diagnosis**
- **Guide treatment by providing antibiotic resistance profile of the infectious agents and helping in the evaluation of the reasons behind treatment failure**

etc

Departament of Infectious Diseases (DDI)

- DDI colaborates in the epidemiological surveillance of infectious diseases, through the activity of several 'Reference Laboratories'
  - Each 'Reference Laboratory' is specialized in a topic, including sexually transmissible infections, respiratory infections, gastrointestinal and food-borne infections, vector-based infections, parasites, fungi, antimicrobial resistance, etc...
- All samples from the national health service suspected of containing an infectious agent are sent to INSa for analysis
  - Samples are analysed by an appropriate 'Reference Laboratory', with some cases being analysed by several different laboratories
- DDI comunicates and collaborates with european and world entities towards the evaluation of biological hazard in the context of public health emergencies
  - Eg. SARS-CoV-2 and Mpox surveillance, among several others

Laboratory surveillance of infectious diseases

Whole-genome sequencing (WGS) as the gold standard typing method

Traditional typing methods

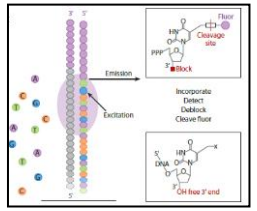
PFGE
MLVA
MLST
PCR assays for detection of virulence genes

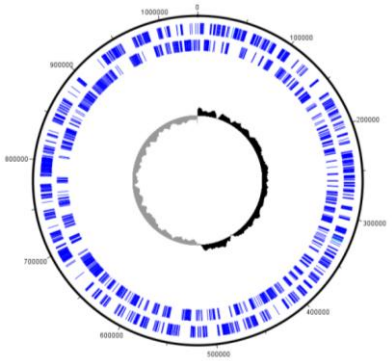
Antibiotic susceptibility tests
Serotyping
(…)

- Difficult standardization
- Time consuming
- Low discriminatory power
(…)

Next-generation sequencing (NGS)

"massive parallel sequencing"



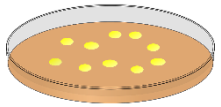


- Capture of the "whole" genome
- High discriminatory power
- Applicable to "all" organisms
- Multiple tests in silico from a single assay
- Cost and time - already competitives
(…)

Laboratory surveillance of infectious diseases

Next-generation sequencing

Isolate
(pure culture)



→ **Whole-genome sequencing**

Complex biological sample
(host + pathogen)

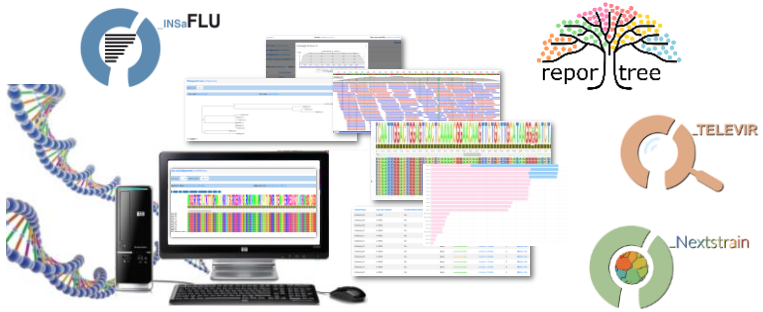
→ **Shotgun sequencing**
(all DNA in sample is sequenced)
→ **Metagenomics for pathogen identification**

→ **Targeted sequencing**
(only the pathogen or specific regions of interest are sequenced)

illumina

Oxford NANOPORE Technologies

NATIONAL REFERENCE LABORATORIES



NEXT-GENERATION SEQUENCING (NGS) ⟷ BIOINFORMATICS / (META)GENOMICS

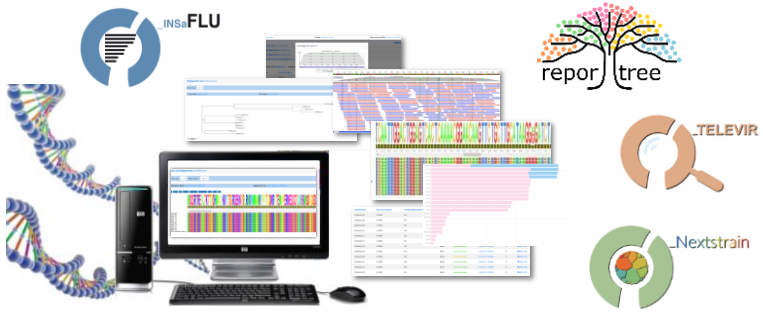
Top from left to right: Sílvia Duarte, Sara Rangel, Camila Fernandes, Luís Vieira, José Ferrão, Catarina Silva. ***Bottom from left to right***: Joana Mendonça, Dina Carpinteiro, Daniela Santos.

Top from left to right: Vítor Borges, Rita Ferreira, Verónica Mixão, João Paulo Gomes, Joana Isidro, João Dourado, Alexandra Nunes. ***Bottom from left to right***: Daniel Sobral, Miguel Pinto, Luís Coelho.

illumina

Oxford NANOPORE Technologies

NATIONAL REFERENCE LABORATORIES



NEXT-GENERATION SEQUENCING (NGS) ⟷ BIOINFORMATICS / (META)GENOMICS

Pathogen's detection
- Response to emerging threats

Monitoring vaccine antigens, antibiotic resistance, etc

Technology transfer
- WGS directly from clinical samples (uncultivable or fastidious pathogens)
- Transition from traditional typing methods to WGS for multiple pathogens

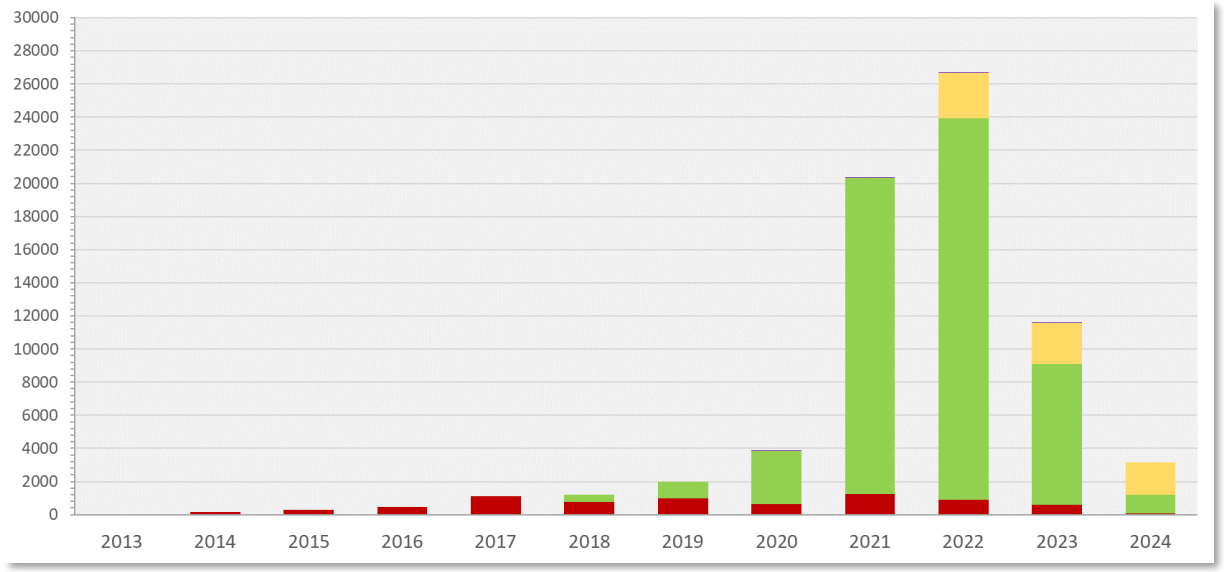
Routine Surveillance / Outbreak Investigation

R&D activities
- Evolutionary adaptation
- Genotype-phenotype associations
- Prediction of antibiotic resistance profiles
- Transcriptomic studies
(...)

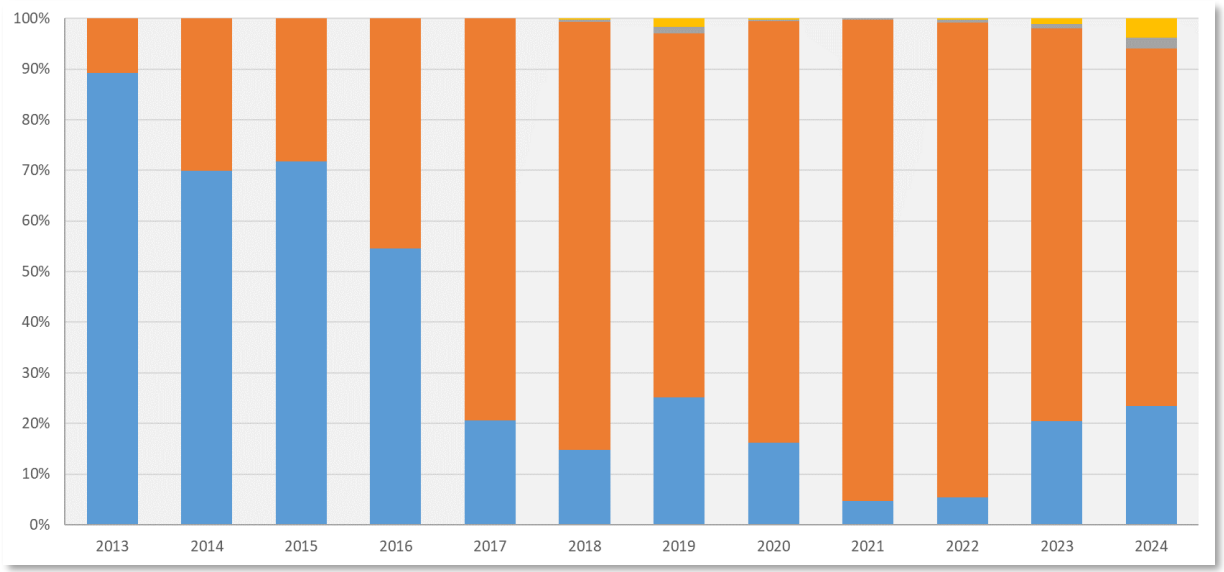
Diagnosis support

Software Development
- Support/Facilitate Laboratory surveillance activities

Total number of samples sequenced over time at INSA



Total number of samples sequenced over time at INSA by purpose



Illumina MiSeq

Illumina NextSeq 550

Illumina NextSeq 2000

Research

Surveillance

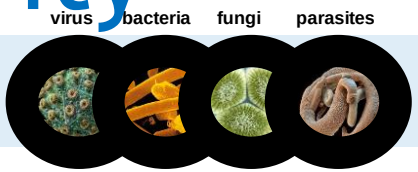
Quality Control assays

Diagnosis

Last update 15/ 05/2024

Examples of GBU activity

"NEXT GENERATION SEQUENCING" (NGS) ⟷ GBU ⟷ NRL



INSaFLU

Genomic surveillance of influenza, SARS-CoV-2, and Mpox vírus, using the INSaFLU bioinformatics plataform

Borges V, Pinheiro M, et al, 2018. Genome Medicine, 10:46.
Borges V, Isidro J, Trovão N et al, 2022, Commun Medicine (…)

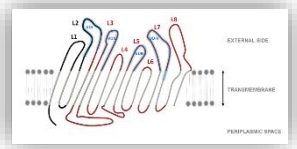
Assisting Diagnosis of Complex Clinical Cases using Metagenomics

TELEVIR

Ferreira et al. (in preparation).

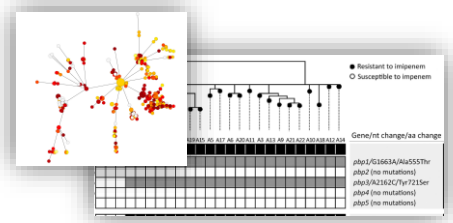
Detection and investigation of outbreaks of:

- Sexually transmitted infections:



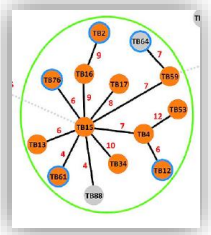
Pinto et al., 2016, Nat Microbiol
Borges et al, 2019, M Genomics, 5
Pinto et al, 2021, M Genomics, 7
Borges et al, 2021, CID, 73

- Food-bourne infections



Llarena et al, 2018, *EFSA*
Isidro et al, 2018; *EID*, 24:741-745
Silveira et al, 2019, IJG
Silveira L, Nunes A et al, 20
Alves F, Nunes A et al., 2022; Front. Microbiol. 13:858310

Monitoring dissemination of multi-resistant *Mycobacterium tuberculosis*



Macedo R, Nunes A et al, 2018, Tuberculosis, 110:44-51
Macedo et al, 2019, Tuberculosis, 115, 81–88.

First sequencing of *Treponema pallidum* (Syphilis) genome directly from clinical samples

Pinto M, Borges V et al. 2016. Nature Microbiol. 2:16190

Investigation of the large outbreak of *Legionella pneumophila*

Borges V, Nunes A et al. 2016. *Sci Reports*; 6:26261
Correia AM et al. 2016. *N Engl J Med*; 374: 497-8.

(…)

INSaFLU-TELEVIR platform (https://insaflu.insa.pt)

SOFTWARE Open Access

INSaFLU: an automated open web-based bioinformatics suite "from-reads" for influenza whole-genome-sequencing-based surveillance

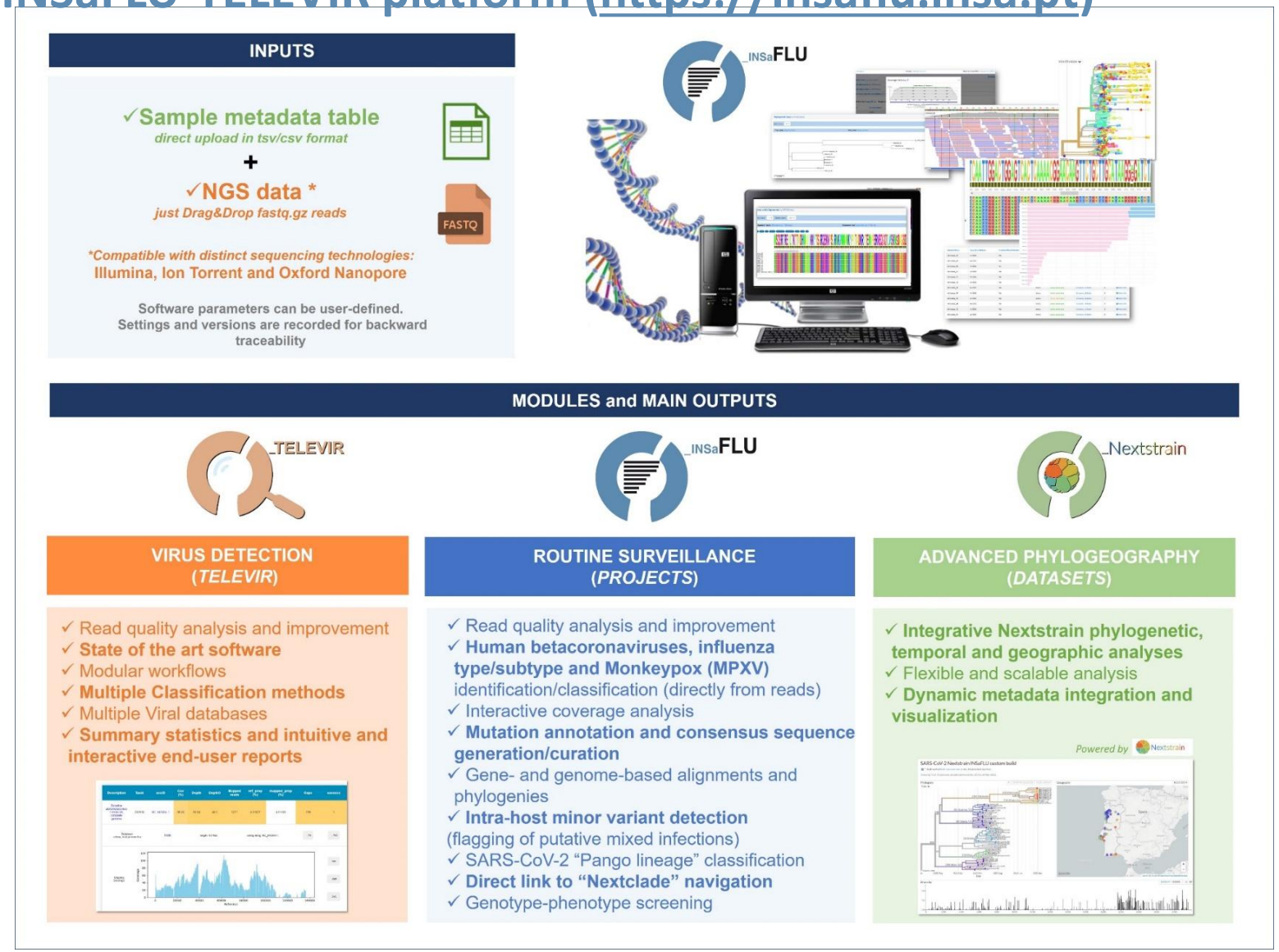
Vítor Borges[1*†], Miguel Pinheiro[2†], Pedro Pechirra[3], Raquel Guiomar[3] and João Paulo Gomes[1]

SOFTWARE Open Access

INSaFLU-TELEVIR: an open web-based bioinformatics suite for viral metagenomic detection and routine genomic surveillance

João Dourado Santos[1], Daniel Sobral[1], Miguel Pinheiro[2], Joana Isidro[1], Carlijn Bogaardt[3], Miguel Pinto[1], Rodrigo Eusébio[1], André Santos[1], Rafael Mamede[4], Daniel L. Horton[3], João Paulo Gomes[1,5], TELEVIR Consortium and Vítor Borges[1*]

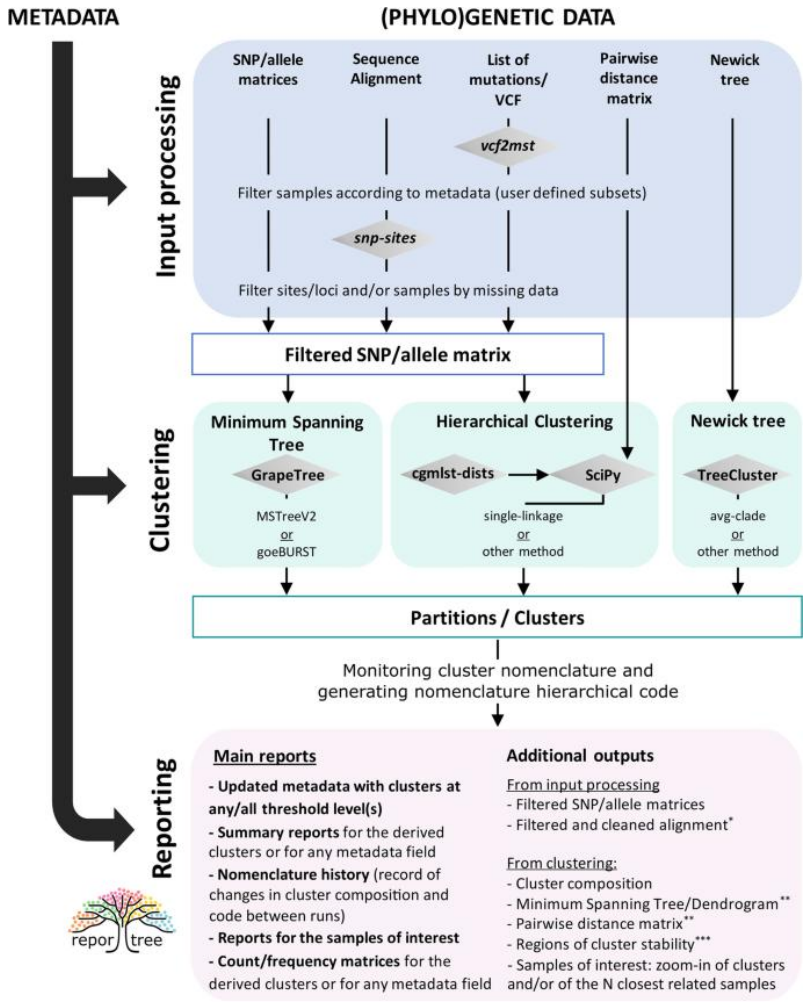
- **Online** (also installable locally)
- **Free** (confidential accounts)
- **User- and surveillance-oriented**
- **Compatible with main sequencing technologies**
- **Multiple features and workflows**



Borges V et al., (2018) Genome Medicine; 10:46
https://doi.org/10.1186/s13073-018-0555-0

Dourado Santos V et al., (2024) Genome Medicine; 16:61
https://doi.org/10.1186/s13073-024-01334-3

ReporTree: a surveillance-oriented tool to strengthen the linkage between genetic clusters and epidemiological data



- Identification of genetic clusters at all resolution levels

  (multiple input formats and clustering methods)

- Surveillance-oriented reports linking to epi data

- Flexible and routine-oriented cluster nomenclature

Outcomes

✓ *Key-role in routine surveillance and outbreak investigation of bacterial and viral pathogens in Portugal*

✓ *Under implementation in WGS systems dispersed across the world*

Mixão et al. (2023)
Genome Medicine 15:43

Contributes to a sustainable and efficient public health genomics-informed pathogen surveillance